

Autoimmune Regulator

A gene called **autoimmune regulator (AIRE)** has been identified as a key player in the prevention of autoimmune diseases. AIRE functions as a transcriptional regulator that promotes the ectopic expression of peripheral tissue-restricted antigens in the thymus. This expression allows developing T cells (thymocytes) to encounter self-antigens during maturation, leading to the elimination of self-reactive T cells through **negative selection**. This process is critical for establishing central tolerance and preventing autoimmunity.

Mutations in the *AIRE* gene result in a rare autosomal recessive disorder known as **autoimmune polyendocrinopathy candidiasis ectodermal dystrophy (APECED)**, also referred to as **autoimmune polyglandular syndrome type 1 (APS-1)**. The classic triad of this syndrome includes: - Chronic mucocutaneous candidiasis - Hypoparathyroidism - Addison disease (primary adrenal insufficiency)

In addition, patients often develop other organ-specific autoimmune conditions such as: - Type 1 diabetes mellitus (DM) - Autoimmune thyroid disease - Gonadal failure - Vitiligo - Alopecia areata - Enamel hypoplasia and nail dystrophy - Pernicious anemia

These manifestations underscore the critical role of AIRE in maintaining immune tolerance by enabling thymic presentation of a broad repertoire of self-antigens.

Cytotoxic T-Lymphocyte Antigen-4

Cytotoxic T-lymphocyte antigen-4 (CTLA-4) is a crucial inhibitory receptor involved in regulating T-cell activation. It is upregulated on T cells

following T-cell receptor (TCR) engagement with antigen presented by antigen-presenting cells (APCs). CTLA-4 competes with the co-stimulatory molecule **CD28** for binding to **CD80 (B7-1)** and **CD86 (B7-2)** on APCs. However, unlike CD28, which promotes T-cell activation, **CTLA-4 delivers an inhibitory signal** that dampens T-cell responses and helps maintain immune homeostasis.

Genetic variations in the *CTLA-4* gene have been associated with an increased risk of several autoimmune diseases. Single nucleotide polymorphisms (SNPs) in both the promoter and coding regions—such as at position **-1722** in the promoter and **+49 A/G** in exon 1—are linked to susceptibility to **systemic lupus erythematosus (SLE)** and **rheumatoid arthritis (RA)**. Although the exact functional impact of these SNPs remains under investigation, they are thought to alter CTLA-4 expression or function.

Notably, a specific SNP (**CT60 G/G**) is associated with reduced levels of a soluble isoform of CTLA-4 (sCTLA-4), which may modulate immune responses by interfering with CD28-mediated co-stimulation. This variant is linked to increased susceptibility to autoimmune endocrinopathies, including: - Graves disease - Autoimmune hypothyroidism - Type 1 DM

These findings highlight the importance of CTLA-4 in controlling peripheral tolerance and its role as a genetic risk factor in autoimmunity.

Programmed Cell Death 1

Programmed cell death 1 (PD-1), encoded by the *PDCD1* gene, is another critical inhibitory receptor expressed on activated T cells, B cells, and myeloid cells. Like CTLA-4, PD-1 functions as a checkpoint molecule that limits immune responses to prevent autoimmunity and maintain peripheral tolerance.

Upon binding to its ligands (**PD-L1** or **PD-L2**), PD-1 transmits inhibitory signals that suppress T-cell proliferation, cytokine production, and survival.

This pathway is essential for preventing excessive immune activation and tissue damage, particularly in peripheral tissues.

A single nucleotide polymorphism (SNP) in the **fourth intron** of the *PDCD1* gene has been associated with susceptibility to autoimmune diseases, including **SLE** and **RA**. This genetic variation may affect PD-1 expression or splicing, potentially impairing its regulatory function and contributing to loss of self-tolerance.

Together, AIRE, CTLA-4, and PD-1 represent key components of immune regulation whose dysfunction—whether through genetic mutation or polymorphism—can lead to the breakdown of tolerance and the development of systemic autoimmune diseases.

Figure Reference

FIGURE 155-1 illustrates the multifactorial mechanisms underlying systemic autoimmune diseases such as SLE. It highlights how genetic factors (e.g., *AIRE*, *CTLA-4*, *PTPN22*, *PDCD1*, *FcγR*, *MHC*) interact with environmental triggers to disrupt immune tolerance. The resulting uncontrolled activation of autoreactive T and B cells leads to autoantibody production and tissue injury.